



SOP-20-880 Smart Operating Panel

Quick Start Guide



19012490A00

1. Overview

The SOP-20-880 smart operating panel is a new generation of commissioning assistant of Inovance that supports HE and MD880 series products.

SOP-20-880 has a wide power supply range and LCD display. It supports multiple buses and applies to the single-motor/multi-motor drive system. It provides such functions as parameter setting, state monitoring, parameter copy, fault analysis, program download, USB relay, and mass storage.

2 Nameplate and Model

SOP-20-880

① Product series SOP: Smart operating panel	③ Product type 880: 880 and HE series supported
② Product type 20: Operating panel series	

MODEL: SOP-20-880
S/N: 01610168VG100099

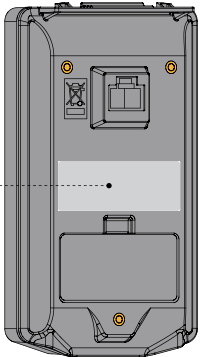


Figure 1 Nameplate

3 Dimension Drawing

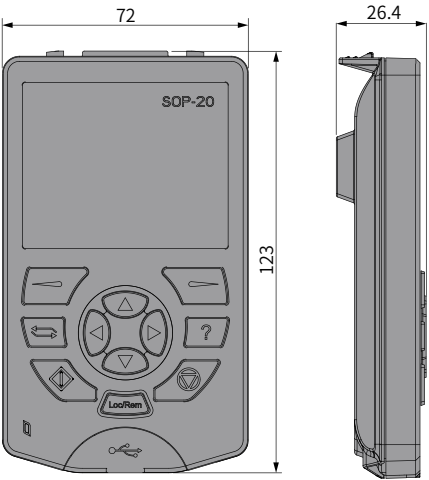


Figure 2 Dimension drawing of SOP-20-880 (unit:mm)

4. Components of SOP-20-880

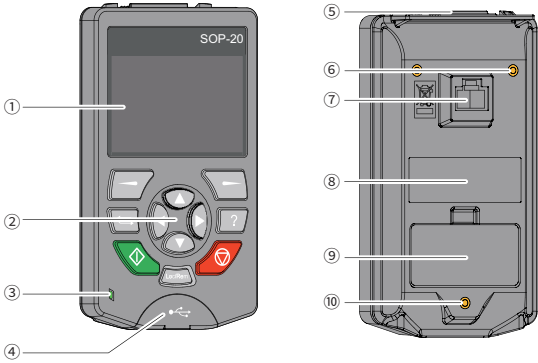


Figure 3 Components of SOP-20-880

Table 1 Interface description

No.	Name	Description
①	Display	-
②	Key	-
③	Indicator	-
④	USB terminal	Used to connect SOP-20-880 to a PC with an USB cable for data exchange. After being connected to the PC, SOP-20-880 can work in relay mode as a PC tool (InoDriveStudio) for data communication with the drive, or work in the mass storage mode to be used for file operations on the SD card.
⑤	Snap-fit joint	The snap-fit joint on the stop is used to fix SOP-20-880 to the base.
⑥	Fixing screw	SOP-20-880 can be installed to the cabinet door by using the installation kit. The IP rating is IP20. When the system complies with IP54, you may select the installation kit with higher IP rating (IP54). Next, secure SOP-20-880 with fixing screws.
⑦	RJ45 interface	Used to connect SOP-20-880 to the drive. Mechanical connection is performed through the snap-fit joint on the top.
⑧	Nameplate	The nameplate displays the product model and serial number.
⑨	Battery cover	The battery and SD card slot are beneath the cover. The battery provides power to the real-time clock. The SD card saves factory parameters and real-time data of the drive, which helps in fault diagnosis and analysis. The SD card can also be used to save user programs, update firmware, and burn the device programs.
⑩	Fixing screw	-

5 Introduction To the Interface

SOP-20-880 adopts the monocolour LCD with white backlight and a resolution of 240x160 pixels.

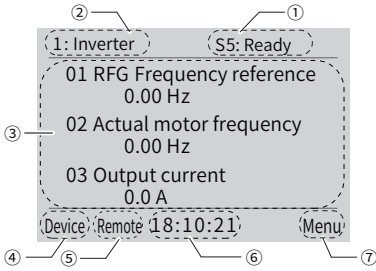


Figure 4 Example of SOP-20-880 interface

Table 2 Interface

No.	Display	Description
①	Device status and fault information	Displays the operating state of current device.
②	Current device information	1: Current device station number; Inverter:Currentdevice name
③	Content	Displayed the monitored data in real time.
④	Device list	Used to enter the device list of the current network.
⑤	Local/Remote	Local: SOP-20-880 has the control permission. Remote: SOP-20-880 has no control permission. No display: This function is not available or the equipment is connected improperly.
⑥	Clock	Displays current time.
⑦	Menu	Used to access the function menu options of current device.

6 Introduction to Keys

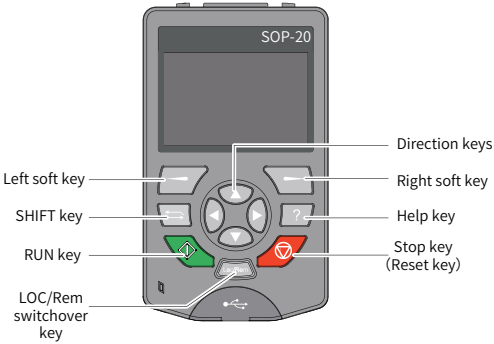


Figure 5 Position of the keys

Table 3 Name and function

Key	Name	Function
	Left soft key	Used for exit or cancellation.
	Right soft key	Used for selection or confirmation.
	SHIFT key	Used to quickly enter the device list. In some interfaces, you are not allowed to switch the devices. In this case, this key is invalid.
	Help key	Used to open the help page. The help page depends on the context, which means the contents of the help page is related to the menu or view.
	RUN key	Used to start the equipment in the local control mode.

Key	Name	Function
	Stop key	Used to stop the equipment in the local control mode. When the device is in the fault state, use this key to reset the fault.
	Loc/Rem switchover key	Used to shift between local and remote control.
	Direction keys	The up and down keys are used to select options in a displayed menu and list, scroll up and down a text page, and adjust a value (such as setting time, entering a password, or changing a parameter value). The left and right keys are used to move the cursor left and right.

Table 4 Description of the shortcut keys

Shortcut key	Available in	Function
Left+Up	Any view	Used to adjust the backlight brightness.
Left+Down	Any view	
Right+Up	Any view	Used to adjust the contrast.
Right+Down	Any view	
Left+Help	Parameter list view	Used to add the parameter where the cursor is directed at to the shortcut interface.
Right+Help	Parameter list view	Used to add the parameter where the cursor is directed at to the startup interface.
Left+Help	Shortcut view	Used to delete the parameter from the shortcut commissioning interface.
Right+Help	Shortcut view	Used to delete all the parameters from the shortcut interface.
Left+Help	Start interface view	Used to delete the parameter from the startup interface.
Right+Help	Start interface view	Used to clear all parameters in the start interface.

7 Indicator Status

The status indicator on the SOP-20-880 is used to indicate whether a fault or alarm exists.

Table 5 Indicator status

Symptom	Example	Meaning
Solid green		The equipment is running properly.
Flashing in green		The data is being transferred between the SOP-20-880 and PC with USB.
Solid red		Communication between the SOP-20-880 and the equipment is normal, but the equipment is faulty.
Flashing in red		Communication between the SOP-20-880 and the equipment has been interrupted.

8 General Specifications

No.	Type	Operation	Storage	Transportation
1	Ambient temperature	-20°C to +55°C	-40°C to +70°C	-40°C to +70°C
2	Attitude	< 4000 m	-	-
3	Relative humidity	95% (without condensation)		
4	Pollution degree	3C3 (IEC 60721-3-3, IEC 60721-3-2, and IEC 60721-1-3-1)		
5	Vibration	61800-5-1 ed 2 EN 60082-2-6 test Fc (1 g)	-	2M3 class (according to EN 60082-2-6)
6	Shock	3M4 class (according to EN 60062-2-27)	-	2M2 class (according to EN 60082-2-27)
7	Free drop	IEC-60068-2-32 Drop height: 1 m	-	-

No.	Type	Operation	Storage	Transportation
8	IP rating	Connected to drive equipment: IP54 (with IP54-compliant accessories) Body: IP20		
9	EMC	IEC 61800-3		
10	Material	Disposal: The SOP-20-880 is recyclable. See related international and local laws and regulations for disposal and recycling.		

9. Installing SOP-20-880 To the HCU Cover

1) Connecting SOP-20-880

- ① Place the bottom of SOP-20-880 to the bottom of the slot on the HCU cover.
- ② Turn SOP-20-880 at the rotary point ① and push the upper part ② until you hear a click.

2) Taking out SOP-20-880

- ① Press and hold down the clip ② with the index finger and middle finger to release SOP-20-880.
- ② Pull the upper end of SOP-20-880 out of the HCU cover slot.

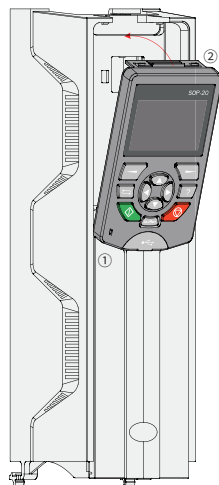


Figure 6 Installing SOP-20-880 to the HCU cover

10. Installing SOP-20-880 To the Cabinet Door

When SOP-20-880 is installed to the cabinet door, secure the base of the panel to the cabinet door with the four screws shown by ③ in the following figure, and then install SOP-20-880 to the base. For details on mounting the base, see SOP-20-880 Operating Panel Base Installation Guide.

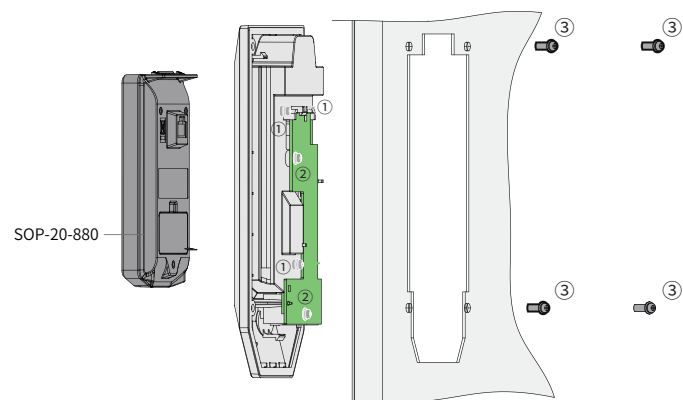


Figure 7 Installing base to the door plate

As shown in the preceding figure, ① represents the screws used to fix SOP-20-880, which are used with IP54 protection. ② represents screws used to fix the PCB, with the grounding function provided. ③ represents screws used to fix the base to the cabinet door.

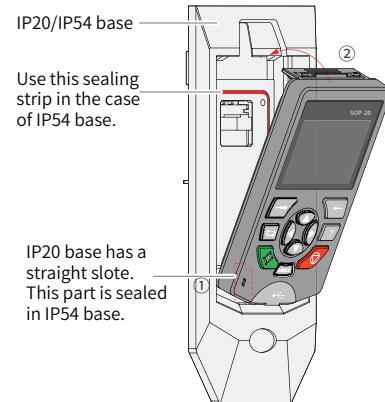


Figure 8 Installing SOP-20-880 to the base



NOTE

External operating panel is not supported when IP54 installation kit is used.

11. Connecting SOP-20-880 to PC

The USB mini-B port on the SOP-20-880 is used to connect the PC. The USB cable length cannot exceed 3 m. The USB2.0 communication protocol is used between the SOP-20-880 and the PC.

- ① Open the cover of the USB connection terminal.

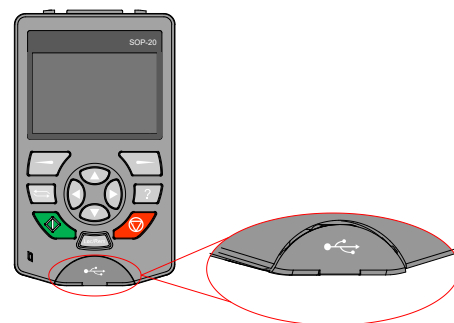


Figure 9 Connecting SOP-20-880 to PC (step 1)

- ② Insert the USB cable and connect the cable to the PC. It is recommended to use cables with ferrite rings.

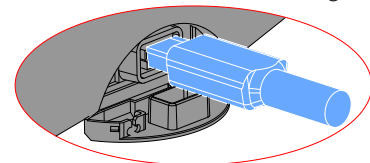


Figure 10 Connecting SOP-20-880 to PC (step 2)



NOTE

USB relay mode and USB mass storage mode are available when the USB cable is connected to PC.

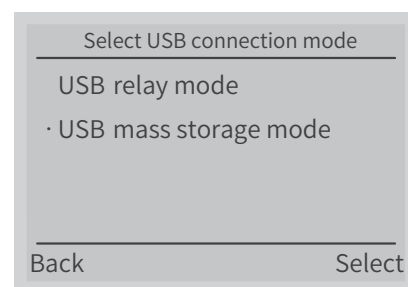


Figure 11 Display of SOP-20-880 after being connected to PC

In USB relay mode, SOP-20-880 is used for interaction between the PC tool and the drive.

After selecting the USB mass storage mode, SOP-20-880 can be used as a memory.

12. Replacing the Battery

- Step 1: Use a tool or fingers to open the battery cover.

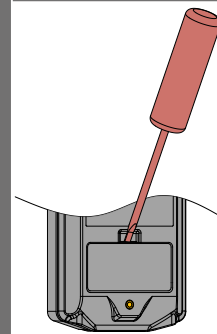


Figure 12 Battery replacement step 1

- Step 2: Take out the battery with tweezers or a slotted screwdriver.

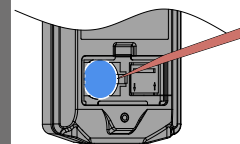


Figure 13 Battery replacement step 2

- Step 3: Place the battery into the clip (side A) first and then press the other end (side B) of the battery.

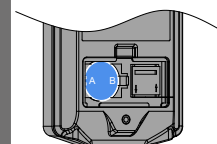


Figure 14 Battery replacement step 3

- Step 4: Put the battery cover back.

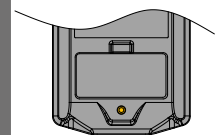


Figure 15 Battery replacement step 4

13. Replacing the SD Card

Hot swapping is not supported by the SD card. The SD card can be replaced only after the SOP-20-880 is powered off.

- Step 1: Use a tool or fingers to open the battery cover.

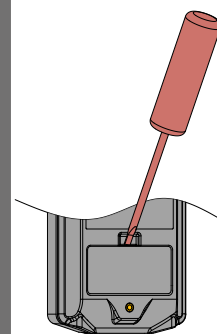


Figure 16 SD card replacement step 1

- Step 2: Push the metal cover towards the arrow direction (downwards) to translate the metal cover to unlock the card holder and take out SD card.

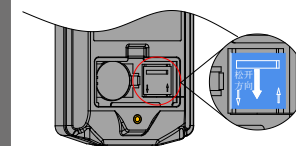


Figure 17 SD card replacement step 2

- Step 3: Lift the metal cover and place a new SD card.

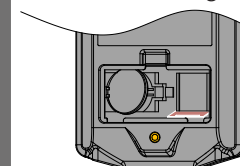


Figure 18 SD card replacement step 3

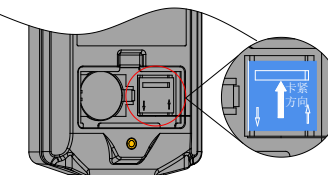


Figure 19 SD card replacement step 4

- Step 5: Put the battery cover back.

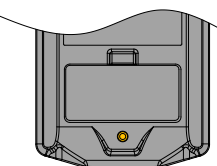


Figure 20 SD card replacement step 5

INOVANCE Warranty Agreement

Inovance provides an 18-month free warranty (subject to information indicated by the barcode on the equipment) to the equipment itself from the date of manufacturing for the failure or damage under normal use conditions.

Within the warranty period, maintenance will be charged for the following damage:

- 1) Damage caused by improper use and unauthorized disassembly, repair, or retrofit;
- 2) Damage caused by fire, flood, abnormal voltage, other disasters, and secondary disasters;
- 3) Hardware damage caused by equipment fall-off or that caused during transportation;
- 4) Damage caused by failure to operate the product according to the user guide provided by Inovance;
- 5) Faults and damages caused by external factors, such as peripheral devices;

If the equipment is faulty or damaged, fill in the Warranty Card correctly.

Maintenance is charged according to the latest Maintenance Price List of Inovance.

The Product Warranty Card is not re-issued. Keep the card and present it to the maintenance personnel during maintenance.

For any question during service, contact Inovance agent or Inovance.

Customers purchasing the product are assumed to agree on terms and conditions of this warranty agreement.

INOVANCE Product Warranty Card

Customer Information	Company address:	
	Company name:	Contact:
	P.C.:	Tel.:
Product Information	Product model:	
	Product barcode (attached here):	
	Agent name:	
Fault information	(Service date and description):	
	Served by:	

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